

Kalpiti Swami

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SUMMARY

Cloud Engineer based in Toronto. AWS Certified Solutions Architect. Co-op at the Government of Ontario supporting 911 ambulance dispatch infrastructure. Built a production-grade EKS GitOps platform with zero static credentials end to end. 1st place AWS Summit JAM 2025, 2nd place AWS User Group JAM 2026.

EXPERIENCE

Information Technology Analyst, DevOps & Infrastructure (Co-op)

Jan. 2026 to Apr. 2026

Ministry of Health (MPBSDP), Government of Ontario

North York, ON

- Reduced Linux/Ubuntu OS provisioning time by **80% across 100+ VMs at 21 sites** by engineering a parallel-safe Ansible playbook with per-VM OVA templates and cloud-init, cutting per-batch runtime from **50 to 5 minutes**.
- Authored PowerShell automation for a **150-machine** Citrix VDA upgrade, compressing a ~**190-hour** manual process to **12 hours** of unattended runtime, a **93% reduction** in manual effort and time.
- Automated device and circuit provisioning in **NetBox using Python** to update router records and create switch entries across **production network infrastructure**, with full documentation maintained.
- Coordinated **100+** infrastructure change requests spanning network patching, Windows patching, and data centre maintenance across **21 sites** through CAB approval in the eSMT portal, achieving a **100% approval rate**.

PROJECTS

Fully Automated EKS GitOps Platform - Zero Static Credentials | [Infra GitHub](#) | [WebApp GitHub](#) | [Dev.to Blog](#)

- Hardened **CI/CD supply chain security** by pinning GitHub Actions to **immutable commit SHAs**, enforcing **Trivy container scanning** and **Checkov IaC validation** on every pull request, blocking **CRITICAL severity findings** before image publication to ECR.
- Eliminated **long-lived AWS credentials** across CI/CD and runtime by implementing **GitHub Actions OIDC federation**, per-workload **IRSA scoped roles**, and **External Secrets Operator** syncing secrets dynamically from Secrets Manager.
- Automated full EKS environment standup from **zero to live in under 20 minutes** by engineering a sequenced pipeline covering Terraform provisioning, **Trivy-scanned ECR builds**, **ArgoCD Helm bootstrap**, and deployment validation.
- Resolved **VPC deletion deadlocks** caused by Kubernetes-managed AWS resources outside Terraform state by implementing **ArgoCD cascade deletion finalizers** and layered destroy sequencing, achieving **fully automated environment teardown**.
- Deployed full-stack observability on EKS by provisioning **kube-prometheus-stack**, **Grafana dashboards** via ArgoCD GitOps, and **8 CloudWatch Alarms** with SNS notifications.

High-Availability AWS Infrastructure with Centralized Observability | [GitHub](#) | [Medium Pt.1](#) | [Medium Pt.2](#)

- Diagnosed and resolved **CloudWatch Agent failure** on ASG-launched EC2s by identifying **IMDSv1 incompatibility** and replacing metadata calls with **IMDSv2 token-based authentication** in user-data.sh, restoring full observability.
- Enforced **tier-to-tier traffic isolation** by implementing layered **security group chaining** across ALB, ASG, and RDS layers, eliminating direct public exposure of application and database tiers.
- Built **modular Terraform infrastructure** with remote state management and a **GitHub Actions CI/CD pipeline** gating all changes behind **manual approval on merge**, following GitOps-style deployment workflows.

TECHNICAL SKILLS

Cloud & DevOps: AWS (EKS, EC2, ECR, VPC, IAM, S3, CloudWatch, SNS, Secrets Manager), Microsoft Azure (coursework), cloud infrastructure, Terraform, ArgoCD, Helm, GitOps

Containers & Networking: Docker, Kubernetes, HPA, reliability, distributed systems, ALB Ingress, RBAC, PersistentVolumes, StorageClass, TCP/IP, DNS, HTTP, NAT Gateway

Security & Observability: IRSA, External Secrets Operator, IMDSv2, least privilege, encryption at rest and in transit, log analysis, Grafana, Prometheus, CloudWatch dashboards

Scripting, Version Control & CI/CD: Bash, Python, Linux (Ubuntu, Amazon Linux), Git, GitHub Actions, troubleshooting

EDUCATION

Postgraduate Certificates in Cloud Architecture & Cybersecurity, Seneca Polytechnic

2024 to 2026 | 4.0 GPA

Coursework: AWS, Microsoft Azure, Terraform, Docker, Kubernetes, networking (TCP/IP, DNS), scripting, security principles, IT infrastructure.

Bachelor of Computer Applications, Indraprastha University New Delhi, India

2020 to 2023 | 3.8 GPA

CERTIFICATIONS & ACHIEVEMENTS

- AWS Certified Solutions Architect Associate (SAA-C03)**, 2026
- AWS JAM: 1st Place** AWS Summit 2025, **2nd Place** AWS User Group 2026 (AWS-hosted hands-on cloud competitions including AI services like Amazon Bedrock)
- Featured in Terraform Weekly Newsletter (Issue #280):** "From Terraform Spaghetti to a Production-Grade GitOps Platform"
- 4x President's Honour List**, 4.0 GPA across all semesters